Team Name: Defi Detectives

Detecting Illicit Cryptocurrency Transactions with Graph Neural Networks

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Project summary

Over the years, cryptocurrency has seen an increase in adoption which has introduced challenges in detecting illicit financial activities such as money laundering and fraud. Detecting such activities is challenging due to the complex, interconnected nature of blockchain data, where transactions are not independent but linked through flows of funds. Traditional machine learning techniques often treat transactions as independent entities, overlooking the relational structure in these systems limiting the capability to capture patterns seen in graphical structures.

In this project we explore the use of graph-based learning methods for detecting illicit activity in cryptocurrency transaction networks. Blockchain data forms a graph structure, where transactions are connected through flows of funds, making it important to consider relationships between entities rather than treating each transaction independently. We apply Graph Neural Networks (GNNs) to incorporate both transaction-level features and network structure, and compare their performance with standard baseline models. We study the problem through both node classification and link prediction tasks to better understand how structural information contributes to identifying suspicious activity. In addition, we examine the extent to which models trained on one blockchain can generalize to another, providing insight into the transferability of learned patterns. Overall, this project aims to evaluate the benefits of graph-based approaches for fraud detection in decentralized financial systems.

What you will do

In this project we will implement and evaluate both baseline traditional machine learning models as well as graph based models for fraud detection on cryptocurrency data. We will begin by constructing a transaction graph representation of the Elliptic Database, where nodes represent transactions and edges are flow of the funds. For the baseline models, we will implement traditional classification models like XGBoost using transaction level features. For the graph based models, we will implement a GNN architecture model which includes GraphSAGE and Graph Attention Network using PyTorch Geometric framework. The above mentioned node classification models will be trained using supervised cross-entropy loss with mini-batch or full-batch training, and evaluated using metrics such as F1-score and ROC-AUC due to class imbalance. For link prediction, we will construct positive and negative edge samples, use dot-product or MLP-based decoders, and optimize binary cross-entropy loss. We will perform hyperparameter tuning over the number of layers, hidden dimensions, learning rate, and neighborhood aggregation size, and conduct studies to analyze the impact of structural information. Additionally we will explore transfer learning by training models on the Elliptic Bitcoin dataset and evaluating their performance on the Ethereum transaction data from BigQuery to assess the generalizability of the learned patterns across blockchain networks.

Related Work & Papers

Prior work on detecting illicit cryptocurrency transactions has primarily focused solely on traditional machine learning techniques, such as logistic regression, random forest, and SVM, which treat transactions independently and fail to capture the relational structure of blockchain data. Weber et al. [1] were among the first to apply a Graphical structure to blockchain data, for anti-money laundering, on the Elliptic dataset, and achieved a f1 score of 0.628 for illicit transactions. More recently, Asiri and Somasundaram [2] compared GCN against several traditional and deep learning models on the Elliptic dataset. Beyond fraud detection, Kasula et al. [3] demonstrated that GCNs can also be applied more broadly to classify blockchain addresses into categories such as exchanges, personal wallets, and service providers, further highlighting the versatility of graph-based methods for blockchain analytics. These works, along with related explorations in GNN-based blockchain fraud detection [4], collectively demonstrate that GNNs are state of the art and motivate our use of GraphSAGE and GAT on Elliptic dataset.

References:

[1] M. Weber et al., "Anti-money laundering in bitcoin: Experimenting with graph convolutional networks for financial forensics," arXiv:1908.02591, 2019.

[2] A. Asiri and K. Somasundaram, "Graph convolution network for fraud detection in bitcoin transactions," Scientific Reports, vol. 15, art. no. 11076, 2025, doi: 10.1038/s41598-025-95672-w.

[3] V. K. Kasula, A. R. Yadulla, M. Yenugula, B. Konda, and S. Ayyamgari, "Blockchain address classification using graph neural networks for enhanced security and analytics," Discover Artificial Intelligence, vol. 5, p. 285, 2025, doi: 10.1007/s44163-025-00508-1.

[4] W. K. Hotor, "Blockchain-based fraud detection using graph neural networks (GNN's)," Master's thesis, Berlin School of Business and Innovation, 2025, doi: 10.13140/RG.2.2.30773.54241.

Datasets

The dataset we will be using for this study is called Elliptic Dataset which maps bitcoin transaction to real entities belonging to licit categories (exchanges, wallet providers, etc.) versus illicit ones (scams, malware, terrorist organizations, etc.). This dataset consists of ~203,000 Bitcoin transactions, ~234,000 transaction edges, 166 transaction features and 49 temporal steps. This makes it well-suited for graph-based learning, since it provides both transaction attributes and the network structure connecting them. We chose this dataset because it supports both node classification and link prediction tasks..

Link to dataset: https://www.kaggle.com/datasets/ellipticco/elliptic-data-set/data